

**Orbital Upcycling: Transforming Space Debris into Valuable Space-Based
Resources**

A focus on the Collection of Space Debris

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DECLARATION

This report entitled

Orbital Upcycling: Transforming Space Debris into Valuable Space-Based Resources
A focus on the Collection of Space Debris

Was composed by me and is based on my own work. Where the work of the others has been used, it is fully acknowledged in the text and in captions to table illustrations. This report has not been submitted for any other qualification.

Name: *Mitansh V Choksi*

Signed:

Date:

Mitansh V Choksi
Date: April 2025
Aerospace Engineering

Abstract

A work in progress. Possibly the last thing I do on this report

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List of Symbols and Acronyms

AoP	=	Argument of Periapsis	RM	=	Rocket Mission Related Object
CONOPs	=	Concept of Operations	RS	=	Rocket Shield
GEO	=	Geostationary Earth Orbit	t_d	=	diameter (no units)
GPS	=	Global Positioning System	t_s	=	ultra-thin (no units)
IADC	=	Inter-Agency Space Debris Coordination Committee	UI	=	Unidentified
ISS	=	International Space Station	WS	=	Whipple Shield
LEO	=	Low Earth Orbit	RD	=	Rocket Debris
MASTER	=	Meteoroid And Space Debris Terrestrial Environment Reference	RF	=	Rocket Fragmentation Debris
PD	=	Payload Debris	PL	=	Payload
PF	=	Payload Fragmentation Debris	PM	=	Payload Mission Related Object
PL	=	Payload	RB	=	Rocket Body
PM	=	Payload Mission Related Object	SMA	=	Semi-Major Axis
RAAN	=	Right Ascension of Ascending Node	SSPS	=	Space Solar Power Systems
JAXA	=	Japan Aerospace Exploration Agency	BLE	=	Ballistic Equations
MTBG	=	Background Meteoroids	Placeholder	=	Placeholder

Notations [1, From Christiansen]

C	=	speed of sound of target (km/s)	m	=	areal density (g/cm ²)
d	=	projectile diameter (cm)	M	=	projectile mass (g)
d_c	=	critical projectile diameter (cm)	P	=	penetration depth (cm)
E	=	elastic modulus in tension (MPa)	S	=	overall spacing between outer bumper and rear wall (cm)
ρ	=	density (g/cm ³)	σ	=	rear wall yield stress (ksi)
H	=	Brinell hardness of target (BHN)	σ'	=	normalized rear wall yield stress (unitless)
t	=	thickness (cm)	θ	=	impact angle from normal to surface (deg)
V	=	projectile velocity (km/s)	V_n	=	normal component of impact velocity (km/s)
c_b	=	0.15 (unitless)	c_{N-K}	=	0.23 (unitless)
c_w	=	8.8 (s/km)	σ'_h	=	$\frac{\sigma_w}{40}$ (unitless)

Subscripts [1, From Christiansen]

b = all bumpers
 t = target

p = projectile
 w = rear wall

Acknowledgements

Firstly, I would like to take a moment to express my deepest gratitude to my supervisor Dr. Angadh Nanjangud for his support, encouragement and patience through this project. There have been times where I was quite lost with the direction of the project and without guidance from Dr. Nanjangud this report would not have reached this far, you have not only equipped me with skills to become a good scholar but also knowledge I can take into my future career, for this I am extremely grateful.

I want to say a massive thanks to my incredible team, WALL-S: Noah Leigh, Emma Nicolai, Joshua Amar and Damian Torres. There could not have been a better team, the collaboration, creativity and support was quite literally out of this world. Each contribution, insight and commitment have elevated the quality of the project and undoubtedly this report. I sincerely hope that the team can come back together for another project in the future.

Under Development, but it's a start...

Chapter 1

Introduction

1.1 An introduction to Space debris and it's growing concern

SPACE! Mankind has been in awe of the heavens above since the beginning of human history. Since the the launch of Sputnik, space has been exploited: commercial satellites for communication and entertainment; satellites for oceanology, meteorology, defence; and telescopes to broaden the understanding of the universe.

This exploitation comes at a cost, not only a fiscal cost but more importantly an environmental cost. The key question that failed to be raised with significance is "What happens to all these sattellites at their end of life?". Until recently, the answer is nothing. Most defunct satellites are left in space. While this was a cost effective and appropriate decision when space exploration began with the launch of 19590 satellites launched in earth orbit² since 1957, it has become a growing concern. With the frequency of satellites increasing in space, there is a higher probability that they will collide and produce debris which orbit in space at much higher velocities. The space debris is likely to create a chain of collisions which results in the creation of "a belt of debris surrounding the earth" as theorised by Donald Kessler.³

The Kessler Syndrome is becoming a significant issue in recent times. An anti-satellite test caused the deliberate "destruction of the Chinese Fengyun-1C in 2007 and the collision of an American and Russian spacecraft in 2009 have increased population of debris in LEO by 70%".⁴ The Inter-Agency Space Debris Coordination Committee (IADC) have reached the conclusion that there is more debris being produced in Low Earth Orbit (LEO) than leaving it, making LEO unstable,⁵ additionally "the IADC and other studies have demonstrated that accidental collisions will drive the future space debris population increase".⁶ [2, The European Space Agency (ESA)] estimates that there are 1.1 million objects in the 1 cm to 10 cm range which are difficult to track. It is these objects which cause the greatest damage to spacecraft, as the spacecraft doesn't have enough time to detect and manoeuvre against small debris.

Recognising that space must be treated as any finite resource on earth with genuine attempts at sustainability, this project is motivatd to solving the issue of small, untrackable debris which poses the greatest threat to current and future space missions.

1.2 Project Aims and Work Streams

The primary aim of the group project is to develop a comprehensive, sustainable solution to solve the problem of remediating small debris which poses the greatest threat to space missions. Out of the various orbits in space, LEO has been identified as the most critical and thus forms the focus of the study. At the time of writing, there are no binding laws for the post-mission disposal of satellites and spent rocket stages, there are a number of treaties and the most recent IADC lays out two possible end-of-life plans for spacecraft in LEO:

- (1) A direct re-entry
- (2) Manoeuvre into a decay orbit as short as possible but not longer than 25 years

The lack of strict guidelines further emphasises the requirement for a mission to plan for the inevitable "littering" of space. Additionally there are concerns with the re-entry proposal. A study by [7, Ferreira et. al] has revealed that the aluminium in space components produce aluminium oxide nanoparticles which act as catalysts for the chlorine activation leading to the degradation of the ozone. The team is aiming to develop a solution which doesn't simply de-orbit any remediated debris, but to capture it, transport it to a local space station for re-manufacturing. The target debris is 1 cm to 10 cm as it poses the greatest threat⁸ and it is an untouched niche. It is expected that the produced products can be manufactured in to custom repair parts for satellites in space.

This paper focuses specifically on the capture of debris, investigating possible remedial methods and then adapting them to capture our target space debris. An analysis is conducted in the feasibility, costs and performance of the proposed method. The work streams for the project are distributed as shown in table 1.1.

TABLE 1.1. Names of project members and allocated subsystem

Name	Subsystem
Noah David Trevelyan Egerton Leigh	Power and Propulsion
Emma Claudia Nicolai	Manufacturing (Materials and Robotics)
Joshua Antony Ian Amar	Materials Processing of Raw Debris
Damian Cervantes Torres	Attitude and Orbital control
Mitansh V Choksi	Capture and Collection of Space Debris (Structures)

This paper focuses on the capture and collection of the space debris. Investigating current methods of space debris remediation and proposing a solution to capture small space debris, backed by analysis of the proposed method. This is the first part of the project mission, capturing and collecting the space debris. ¹

¹Note: The introduction is based on the front end design report of [9, Choksi (2024)]

Chapter 2

Background Research

2.1 Detailed Literature Review

Fig 2.1¹⁰ shows the increase in debris over time, and in recent years the payload debris (PD) is showing sudden increase, this can be explained by events such as the break up of Intelsat 33e which added approximately 500 pieces of trackable debris.¹¹ This trend is expected to grow, especially with an increase in launches and the development of constellations, this is captured by Fig. 2.2.

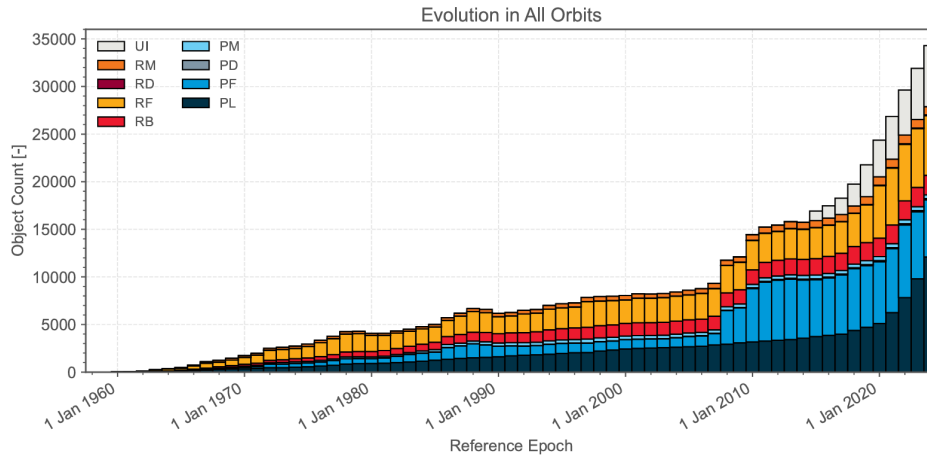


FIGURE 2.1. The number of tracked space objects since 1960s.¹⁰ See Acronyms for definitions.

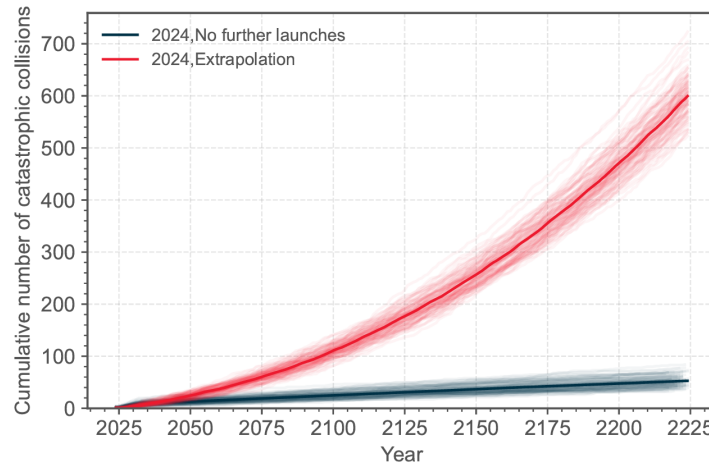


FIGURE 2.2. "The cumulative increase in the number of catastrophic failures extrapolated if no implementation is made to treat the problem of space debris"^{9,10}

In Fig. 2.3, the spatial density of objects (1 cm to 10 cm) can be seen against the altitude for human objects and the background meteoroids (MTBG). It is clear that a significant portion of the total objects are human. This figure was produced using Meteoroid And Space Debris Terrestrial Environment Reference (MASTER) software developed by ESA.

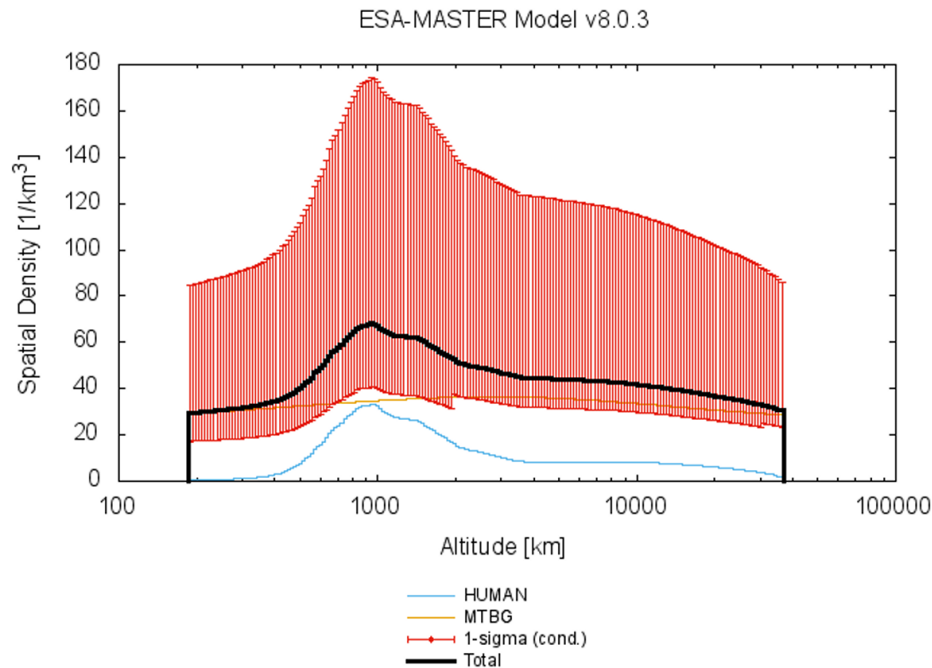


FIGURE 2.3. "A plot of the spatial density of objects (in the 1 cm to 10 cm range) against altitudes ranging from 186 km to 37786 km. Data is from 2016 projected to 2036 using MASTER"⁹

A key issue in the capture of small debris is the spatial density. When a collision or explosion occurs, the debris not only maintains its original momentum but gains an impulse from the explosion which has a different velocity vector to the original spacecraft. The result is a decrease in concentration at the original site as "the right-ascension and declination of the breakup site change over time".⁸ Thus, the spatial density is low along with the probability of locating with the debris. This is an essential point to consider when determining the method of collecting space debris.

It's estimated by [12, Colvin et. al] that there is a benefit of \$23 M in the first year of debris removal and a cumulative benefit of \$7 B over a twenty-five years. It is expected that this number will only increase as the problem deepens and more players enter the market.

2.1.1 Methods of capturing debris

There are numerous methods used for capturing debris such as: "Solar sails; lasers (ground based and space); electrodynamic tethers;¹³ rapid response rockets and recycling into ΔV ^{8,12".⁹ The main remediation methods are summarised in the Table2.1 below.}

TABLE 2.1. A table of proposed remediation methods and their concept of operations

Method Proposed	Concept of Operations (CONOPs)
Sweeper in Orbit	Involves a structure like a shield upon which the debris impacts. The impact of the debris reduces the velocity and promotes re-entry, the debris shatters into smaller pieces which are a lower threat to spacecraft. "Various designs by [14, Takeichi and Tachibana], [12, Colvin et. al], [8, Pulliam] and [15, Werner] have been proposed which follow a similar principle." ⁹
Robotic Arm	A satellite attached with a robotic arm to rendezvous and "tug" The debris to the Earth to de-orbit it. ¹² The major issue is rendezvousing with the debris and matching it's orbital velocity, "rendezvous axis, the tumbling axis and the de-orbiting direction". ¹⁶ The robotic arm can only typically attach to the mounting plate, if it's still intact.
Tether Nets	A net is deployed from a satellite which envelops and entangles the debris in microgravity. [17, Botta] advises a "closing mechanism is used to ensure that the net remains wrapped around the debris." A key concern is the severing of the tether due to debris and using a closing mechanism in microgravity.

2.2 Analysis of the Sweeper Method

From various studies thus far, the sweeper design is an apt choice for debris in the range of 1 cm to 10 cm provided it is modified to suit the parameters of the mission. Since sweepers must be large enough to intercept debris passively, they have a large mass which poses the a big challenge in context of launch costs and vehicles. "Desirable properties include :a lightweight material;⁸ high toughness, ductility and impact resistance."⁹ [18, Cour-Palais and Crews] proposed a concept with "low t_s/d_p " which reduces mass considerably.

In 1947, [19, Whipple] proposed a "bumper" to "dissipate the penetrating power of the meteorite" as a solution to protecting satellites from undesirable impacts. This has commonly been developed into the Whipple Shield (WS), a current standard for protection against debris.

Fig. 2.4 shows a simple diagram of a WS. The debris impacts the bumper, it shatters into many smaller fragments and the debris cloud produced is absorbed inside. The debris impacts the rear wall which is typically thicker and prevents debris from entering the spacecraft/satellite, a larger stand-off distance ensures less damage to the rear wall.

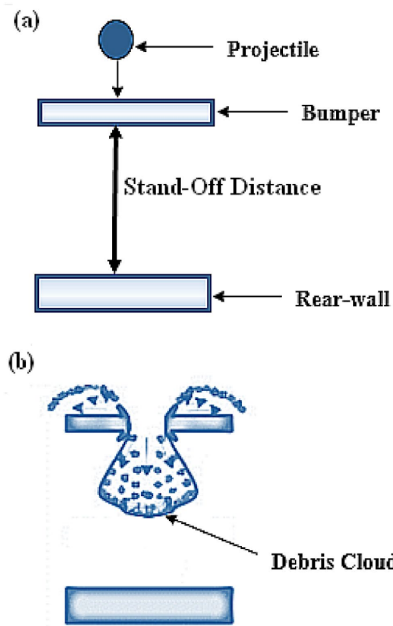


FIGURE 2.4. A diagram of a simple Whipple shield²⁰

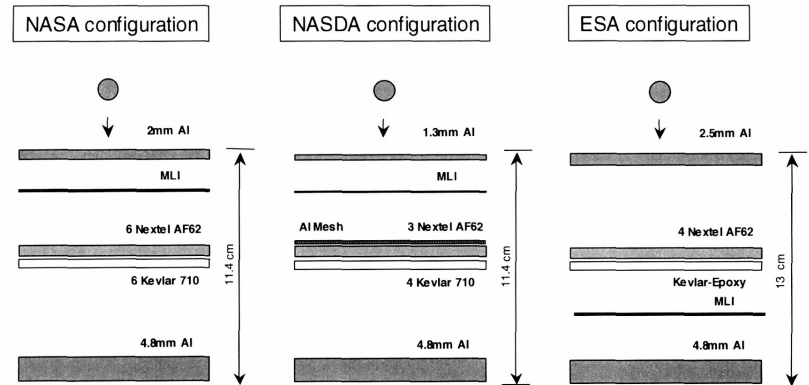


FIGURE 2.5. Whipple Shield Configurations on the ISS.¹

Fig. 2.5 shows the configuration of WSs used on the ISS by various modules designed by different space research organisations. They differ from the basic design in Fig.2.4 as they have more intermediate layers which improve the performance of the shield.

2.2.0.1 Adapting the Sweeper Design for the Project

Based on the design of the WS thus far, there is an opportunity to adapt the design further to suit the requirements of capturing debris. The concept of operations proposes the debris impacting the sweeper structure which absorbs the impact and the debris is embedded in the structure. Then the collected debris and the damaged sweeper parts can be recycled in the "up-cycling" phase of the mission.

"The remainder of this report assesses the possible designs, the materials, costs and associated with this proposition. Research is carried out on configurations of the WS and the effects of a hyper-velocity impact from the debris."⁹

1

¹Note: Chapter 2 is based on the front end design report of [9, Choksi (2024)]

Chapter 3

Proposed Design and Capture Method

3.1 Design of the Sweeper

Before the sweeper it is important to assess the orbital elements (a , e , i , Ω , ω) and the probability of objects it is expected to collide with and the materials involved in the sweeper. This then provides enough information about approximate size, shape, mass and costs of the structure to carry out further analysis.

3.1.1 The target orbit and flux of objects captured

Ideally the target orbit is an orbit with the most debris. MASTER (v8.0.3) is used to determine the initial target orbit for the mission using the 2028 epoch. Using the spatial density tool charts of spatial density against the the altitude and the Right ascension of the ascending node (RAAN) and the declination are shown in Figs. 3.2 and 3.1.

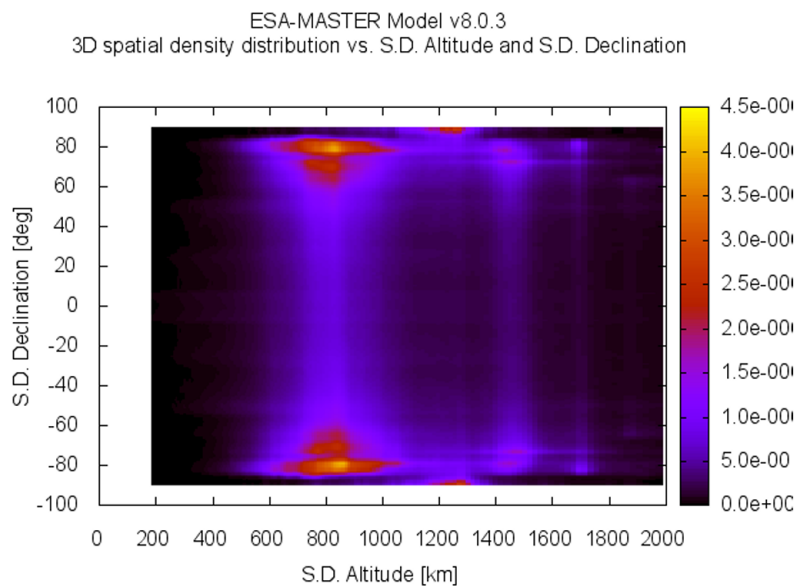


FIGURE 3.1. "A chart comparing the spatial density against the altitude and the declination"⁹

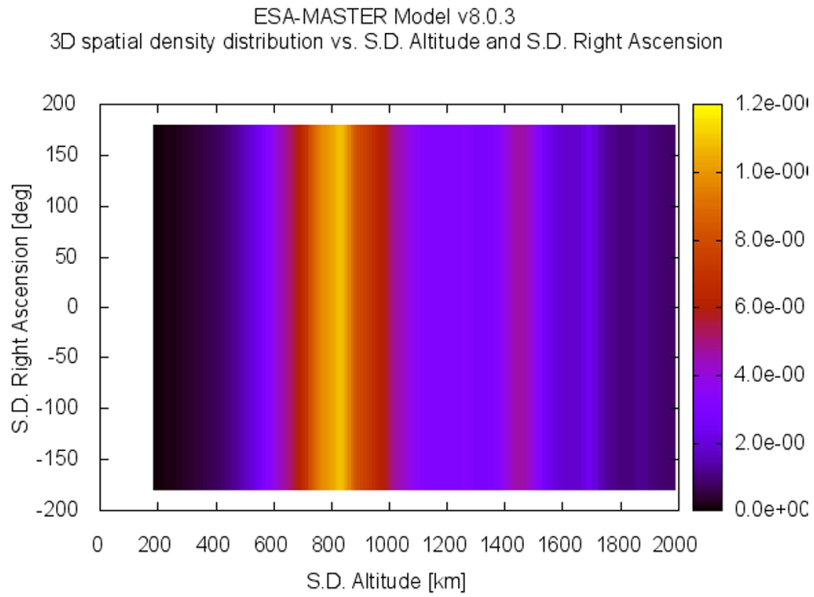


FIGURE 3.2. "A chart comparing the spatial density against the altitude and the RAAN"⁹

From Fig. 3.1 it is seen that there is a higher flux of debris in the $\pm 60^\circ - 85^\circ$ range. Fig. 3.2 shows that varying the RAAN doesn't impact the flux, although both Figures show that the ideal altitude is 700-1000 km. This is consistent with the findings of [21, Obukhov et al.] which show that there is a more debris in the range of $\pm 80^\circ - 100^\circ$. Based on these findings an initial test simulation was run with the orbital elements shown in 3.1, various azimuth and elevation angles were tested, eventually it was settled on 0° for both, this gives a maximum surface area, facing normal to the surface of the earth. Based on this orbit the maximum flux of objects was found to be $2.76\text{E-}4$ ($1/\text{m}^2/\text{yr}$). The number of objects the sweeper collides with can be found using 3.1 which is used in 3.2.

TABLE 3.1. The orbital parameters set for the initial simulation and the predicted flux using MASTER

Epoch	SMA (km)	Eccentricity	Inclination ($^\circ$)	RAAN	AoP	Surface azimuth ($^\circ$)	Surface impact elevation ($^\circ$)
2028	7228	2.697E-4	81.0	30.5681	93.9282	0	0

$$\text{Number of Objects} = \text{Flux} \times \text{Area} \times \text{Time Period} \quad (3.1)$$

3.1.2 The materials involved and size of the sweeper

Fig. 2.5 shows the materials used currently in WSs on the ISS. They primarily consist of aluminium as it has a high strength to weight ratio compared to materials such as titanium. The materials used in the intermediate layers are typically kevlar and nextel due to their high toughness. The properties are shown in Table .1 in Appendix A.

Assuming that the sweeper takes a circular shape and uses Al-6061-T6 as for the bumper and rear wall. "Based on Fig.2.5 let the thicknesses be: bumper: 2 mm; Nextel and Kevlar: 2.5 mm; Rear wall: 4.8 mm. The ideal launch vehicle would be the Falcon Heavy booster which launches 64 tonnes of payload to LEO with a cost of \$1400 per kg.^{22,9}

"Using this; the flux calculated and Eq.(3.1), its possible to calculate, for a passive sweeper, the mass; the number of objects it collides with; the approximate launch costs and number of launches. This is summarised in Table 3.2 below."⁹ This is a simple calculation for an approximation of the values.

TABLE 3.2. For a passive sweeper: the diameters of sweepers, their total mass, the approximate costs, and number of launches⁹

Diameter (m)	Total mass (tonnes)	Number of collisions	Launch costs (USD)	Number of launches
10	2.50	0.02	3.75M	1
100	250	2.16	375M	4
200	1,000	8.66	1.50B	16
300	2,250	19.5	3.38B	35
600	9,000	77.9	13.5B	140
1000	25,000	216	37.5B	390

A diameter of 1000 m would be suitable, but its expensive. Its more suitable if the diameter is 300 m. This doesn't capture as many objects, however the mission uses active detection and interception, hence its expected more objects can be captured.

While this is a large structure for space, structures such as the SSPS by JAXA²³ have been proposed which are larger still. Hence the proposed idea is feasible albeit an engineering challenge. Additionally, if ten sweepers of the above size are launched in orbits with varying inclination in the range of $\pm 80^\circ - 100^\circ$, more debris would be collected simultaneously.

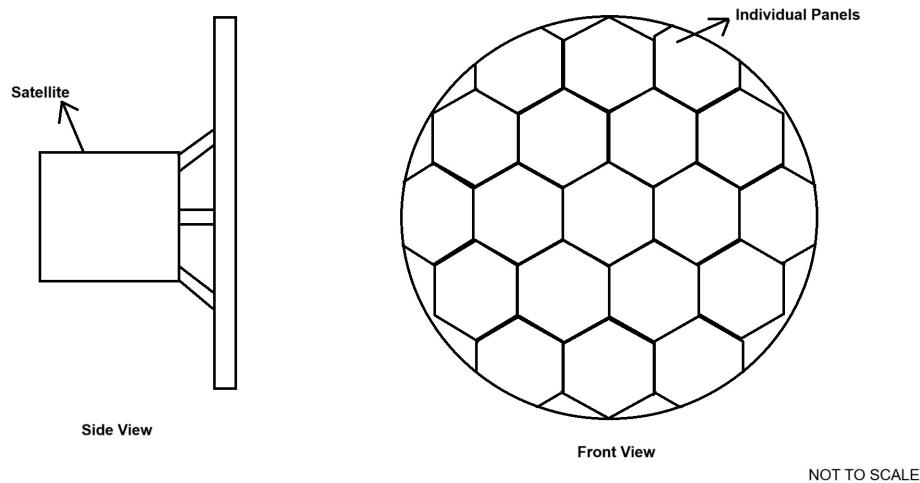


FIGURE 3.3. A sketch of the side and front view of the sweeper

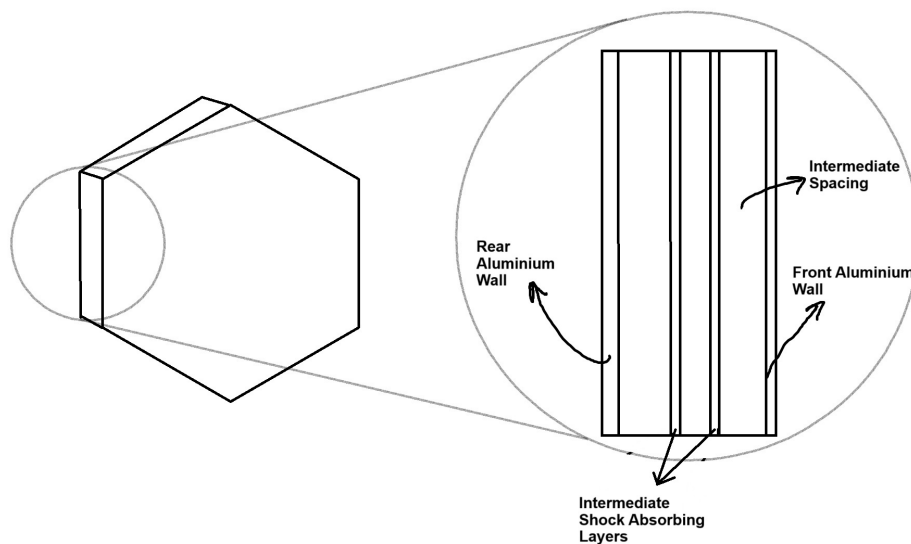


FIGURE 3.4. A sketch of the whipple shield implementation

Sketches have been produced for the final design of the sweeper, these are seen in Fig.3.3 and Fig. 3.4. It consists of many panels which would be assembled in space. This would allow parts of it to be replaced if damaged.

3.2 Capture and Hyper-velocity Impacts

Debris in the targeted range has a velocity in the range 3 - 18 km/s,²⁰ this is defined as hyper-velocity. Debris makes hypervelocity impacts with the shield. The analysis thus far considers generic WS

configurations used on the ISS, designed over 20 years ago. For the remainder of this project the aim to investigate various configurations of the shield and adapt them to capture debris, reduce mass and extend the lifespan. Results thus far will serve as a benchmark for future investigation.

For designing of WS its important to assess the suitability of materials using cratering; design, and performance ballistic equations (BLEs) to test various configurations of spacing before simulating the proposed configuration.

3.2.1 Cratering equations

For $\rho_p/\rho_t < 1.5$,

$$P_\infty = 5.24d^{19/18}H^{-0.25}\left(\frac{\rho_p}{\rho_t}\right)^{1/2}\left(\frac{V_n}{C}\right)^{2/3} \quad (3.2)$$

For $\rho_p/\rho_t \geq 1.5$,

$$P_\infty = 5.24d^{19/18}H^{-0.25}\left(\frac{\rho_p}{\rho_t}\right)^{2/3}(V_n/C)^{2/3} \quad (3.3)$$

To obtain the thickness of the rear wall without spall:

$$t_w = 2.2P_\infty \quad (3.4)$$

3.2.2 Design and Performance ballistic equations

Design and performance equations assist in determining the key parameters of the WS. They differ slightly on each set up, e.g. the Nextel/Kevlar stuffed whipple BLEs differ from multishock shield BLEs. Equations 3.5- 3.7 are the design equations for the Nextel-Whipple stuffed WS. The performance equations vary on various impact velocities and can be found at [1, Christiansen's report]

$$t_b = C_b d \frac{\rho_p}{\rho_b} \quad (3.5)$$

$$m_{\text{Nextel-Kevlar}} = C_{\text{N-K}} d \rho_p \quad (3.6)$$

$$t_w = c_w \left(\frac{(t_b \rho_b + m_{\text{Nextel-Kevlar}})}{(c_o d \rho_p)} \right)^{1.1} M_p V_n (\cos^{1/2} \theta) \rho_w^{-1} S^{-2} \sigma_h^{-1/2} \quad (3.7)$$

Equations 3.2- 3.7¹ will be used next semester for the analysis of suitable configurations of the modified WS on the sweeper. Once a suitable shortlist of configurations is determined, Ansys Autodyn will be used as a numerical solver for the accurate determination of impact and studying the effects on damage to the shield.

3.2.3 Momentum impulse and change in ΔV

3.3 Safety concerns, Risks and Mitigation

The main concern with safety is the interception and collision of larger, functional satellites. This not only destroy the larger satellite in orbit but also produces more debris in the process. This can be mitigated using onboard sensors such as LiDAR or cameras to predict spacecraft trajectories and maneuver. A further risk is the loss of communication with missions control on earth in such a scenario, an ideal system would be equipped with automation and on-board decision making capabilities. Similarly, as the structure has a proposed diameter of 300 m, to avoid collisions with unwanted objects, it should have the ability to fold/reduce the surface area when in close proximity. Another major concern is uncontrolled re-entry. If the system malfunctions, the entire satellite and sweeper could re-enter atmosphere and pose a large threat to life on earth. To mitigate this, a fail-safe system must be built in to ensure the de-orbit process is safe and a large proportion of the structure "burns up" in the atmosphere. A risk with this concept is the production of secondary debris generated when debris fragments on the bumper, this defeats the purpose of the mission and if the fragments are larger than 5 mm then it poses a threat to other spacecraft.⁸ These are some of the major concerns with the concept. Its planned these will be addressed in more depth in the report next semester.

Chapter 4

Developing an efficient structure/Results

4.1 An introduction to the method using BLEs and numerical simulation

4.2 A baseline reference using the ISS NASA/ESA WS Configuration

4.3 Choosing Alternative materials and configurations

Some information on why we are choosing alternative methods (cost, mass, lifespan etc. sustainable manufacture stuff here), then label reference back to the chapter 3 Display the data for each material in tables plus references. This can all go into the appendix. Using proper referencing will ensure things are easy to access.

4.3.1 Choosing X

4.3.2 Choosing Y

4.3.3 Choosing Z

4.4 How the proposed materials affect cost

This includes the launch cost, materials cost, logistics of the whole project

4.5 *Final Design, CAD and Engineering Drawings*

4.6 End of Life Plans

How does the current choice of the materials affect the end of life and sustainability goals

Chapter 5

Discussion

5.1 General Discussion

The main aim of this study is to investigate the issue of small space debris and propose a method of capturing it so that it can be re-used in space for manufacturing as another phase of the overall project. The research and analysis has led to the sweeper method being used for the collection of small space debris. The idea is that the debris will embed in the WS and it can be separated at a later stage of the process. WSs are heavy and expensive to launch using the materials discussed for the ISS in section 3.1.2. The advancements in research in the last 20 years look at the use of open cell metallic foam²⁴ which shows more promise over alternatives such as honeycomb sandwich panels and traditional WSs. Research by [25, Jing et. al] has found the use of an aerogel/fiberglass composite in shielding to be effective and this also improves the thermal insulation. Silica aerogels are promising in the field, although they require additional reinforcement, [26, Sharma et. al] propose the use of polyimide aerogel tiles. There is potential to reduce launch costs with the Starship boosters developed by SpaceX. These would potentially reduce the price of launching payload to \$100 - \$200 per kg.²⁷

5.1.1 The appropriateness of the software used

MASTER has been used to understand the flux of debris in orbits and determining whether the sweeper placed in an orbit will collide with objects in space. Its a reference model used for predictions and results produced for small debris may be less reliable as they are difficult to track and predict. To improve the reliability of the results, its advisable to run the same orbit parameters using the NASA ORDEM model, another reference frame used for understanding debris impacts, this validates the current results. Ansys Autodyn is popular in simulating the effects of hypervelocity impacts and [20, Pai et al.] shows that many studies have used Autodyn as the numerical solver. The analysis in the next semester will involve testing many materials, the software would prove useful for fast analysis. Additionally, it provides the most accurate analysis short of an experimental set up which requires considerable resources.

5.2 End of Life Plans

The sweeper concept can only be used once. The capture of the debris also destroys the sweeper in process. Since its made of panels the logical method would be to recycle the sweeper panel by panel using the aluminium captured from debris and the proposed manufacturing method to replenish the panels in space to allow the sweeper to continue it's mission.

5.3 Individual achievements and how they contribute to the overall mission

My section of the overall project focuses on the collection of the space debris. This study has produced a feasible idea to target space debris in the 1 cm to 10 cm range and identified suitable inclinations and altitudes for the orbits. This contributes to the overall mission directly. Additionally, the analysis of the proposed idea has produced baseline values for the size, proportion and mass of the overall structure. This is useful to Noah for accurately determining the propulsion of the sweeper and to Damian for the AOCS. The analysis on the approximate number of passive collisions also contributes to Joshua's research and aids him to identify the minimum size of the reactant vessel for processing of the debris.

5.4 General Remarks

5.4.1 Appropriateness of the software/method used

5.5 Feasibility/EoL/Future Work

Chapter 6

Conclusions

6.1 Summary and Conclusions

The main conclusion is that the capture of small debris is a difficult task, the most appropriate method is the active space sweeper concept which has been pursued. The debris is captured when it embeds within the sweeper upon impact. It is concluded that the best orbits for the collection of debris cover declination regions of $\pm 60^\circ - 85^\circ$, ideally this would require collection in inclinations of $80^\circ - 100^\circ$. After considering the mass of the structure and the launch costs, it was concluded that a sweeper with a diameter of 300 m provides the best compromise collecting objects versus the size and mass, this would cost approximately \$3.4 B to launch based on the configuration of a Whipple Shield on the ISS. It is expected that there will be 20 desired object collisions passively, the actual number is expected to be higher as the sweeper will be active. To improve the yield of the numbers of objects, it is proposed that 10 sweepers can be launched simultaneously in multiple orbits with the proposed inclination.

6.2 Future work and recommendations

A detailed study will be conducted on the actual capture mechanism through hypervelocity impacts. Suitable advanced materials need to be identified to reduce the mass of the structure and measure the performance against the ISS baseline. This will be achieved using existing BLEs (see 3.2) from research and validating this using simulations using Ansys Autodyn. Additional work must be carried out to assess the thermal stability of the structures, the effect of radiation on materials like aerogels and a FEA of the entire sweeper to identify sites of structural failures. If more time is given then future work can also look at the folding mechanism (3.3)

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Appendix A

TABLE .1. Common materials used in Whipple Shields and their properties

Material	Density (kg/m^3)†	Tensile Strength (MPa)	Young's Modulus (GPa)	Coefficient of thermal expansion ($\mu\text{m/m-K}$)
Al-2024-T3*	2.77E3	483	73.1	23.3
Al-6061-T6*	2.71E3	310	68.9	23.6
Al-2017-T4*	2.79E3	427†	72.4†	23.6†
Nextel 610	3.90E3	2200†	385†	7.1†
Kevlar 149	1.47E3	3400†	180†	-2.5†

* [20, Pai et. al] † [28, ANSYS Edupack]